

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Yellow-billed Cuckoo"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

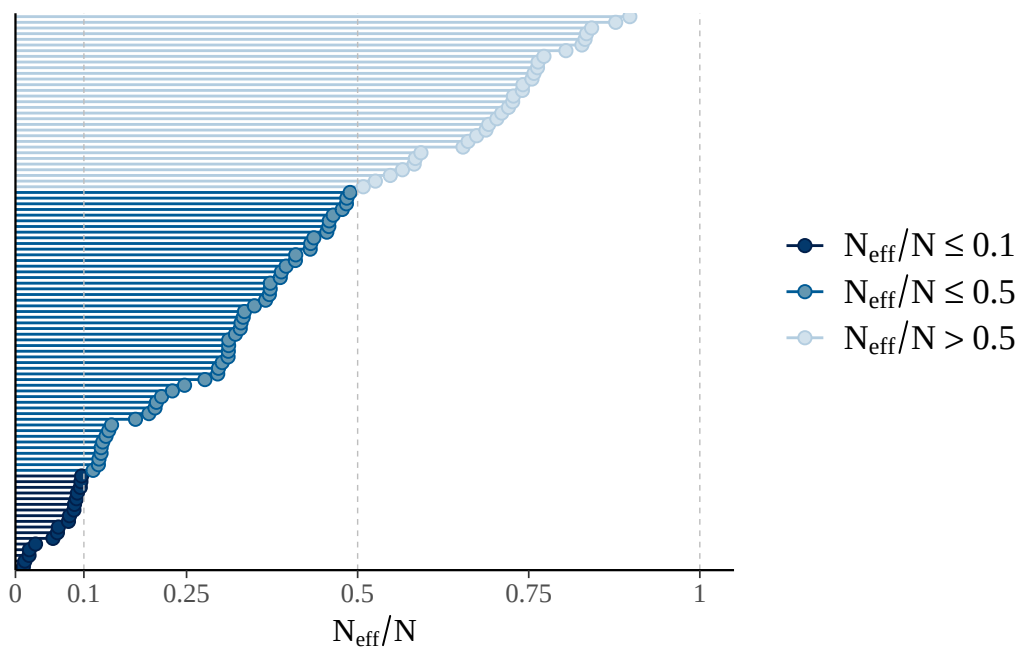
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size too small"
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01159	0.13524	0.37556	0.50064	0.68848	1.52248

[1] "n_eff ratio tail ESS:"

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02032	0.31391	0.49873	0.48390	0.68232	0.89769

[1] "effective sample size in bulk or tail too small"

Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.



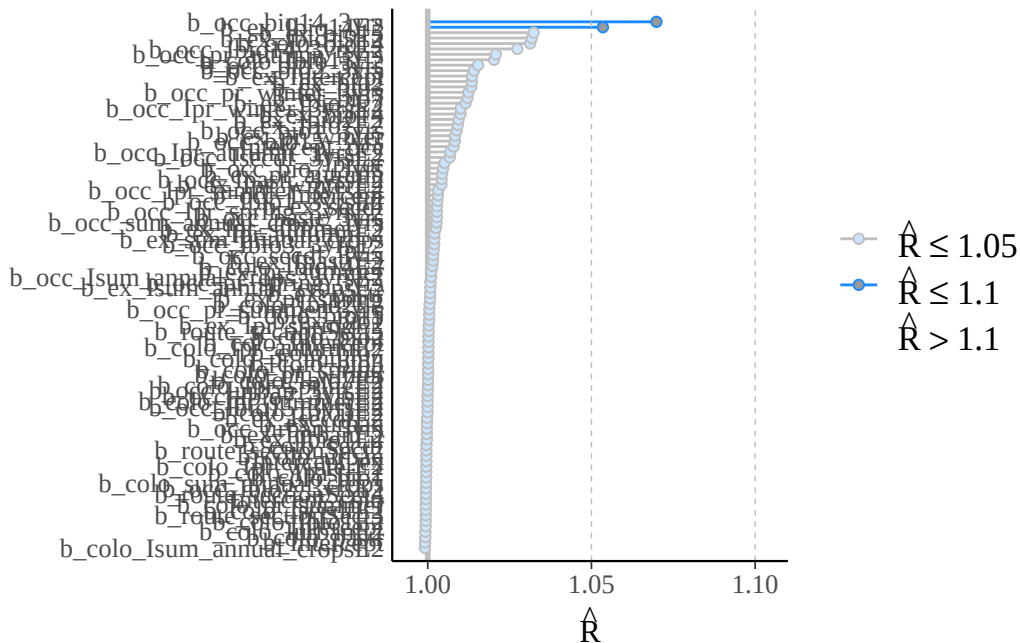


Post-warmup iteration

Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9991	0.9998	1.0008	1.0054	1.0069	1.0699

```
[1] "R-hat > 1.05 for 2 parameters."
```



Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"          "b_occ_bio1_3yrs"
[3] "b_occ_Ibio15_3yrsE2"     "b_occ_Ipr_winter_3yrsE2"
[5] "b_occ_urban_3yrs"        "b_occ_Isum_annual_crops_3yrsE2"
```

```
[1] "b_colo_bio1" "b_colo_pr_spring"
[3] "b_colo_pr_summer" "b_colo_pr_autumn"
[5] "b_colo_lbio1E2" "b_colo_lpr_springE2"
[7] "b_colo_lpr_summerE2" "b_colo_sum_annual_crops"
[9] "b_colo_secdf" "b_colo_urban"
[11] "b_colo_lsum_annual_cropsE2" "b_colo_lsecdfE2"
[13] "b_colo_lurbanE2"
```

[1] "b_ex_bio1" "b_ex_Ibio2E2" "b_ex_Ipr_summerE2"
 [4] "b_ex_urban"

